Advanced SQL
Exercise 6
Date functions

1. Add 6 months to each employee's hire date using DATEADD()

```
SELECT emp_id,
       name,
       hire_date,
       DATEADD(Month, 6, hire_date) AS hire_plus_6_months
FROM Employees;
```

emp_id	name	hire_date	hire_plus_6_months
1	Alice	2020-01-15	2020-07-15
2	Bob	2021-06-10	2021-12-10
3	Charlie	2023-03-22	2023-09-22

2. Use DATEDIFF() to find age in days from dob to today

```
SELECT student_id,
       name,
       DATEDIFF(DAY, dob, '2026-04-17') AS age_in_days
FROM students;
```

Student_id	name	age_in_day
101	Maya	7556
102	Ethan	7800
103	Sienna	7337

3. Find out how many days are left until each event using DATEDIFF().

```
SELECT event_id,
       event_name,
       DATEDIFF(DAY, '2026-04-17', event_date) AS days_remaining
FROM Events;
```

event_id	event_name	days_remaining
1	Seminar	-671
2	Workshop	-228
3	Hackathon	-452

Note:
Exercise done on 17 April 2026, dates have passed, hence the negative.

4. Calculate the number of days between issue_date and due_date.

```
SELECT invoice_id,
       issue_date,
       due_date,
       DATEDIFF(Day, issue_date, due_date) As days_between
FROM Invoices.
```

invoice_id	issue_date	due_date	days_between
501	2025-03-10	2025-03-25	15
502	2025-04-01	2025-04-15	14
503	2025-04-10	2025-04-20	10

5. Format start_date as 'Month YYYY' using TO_CHAR().

```
SELECT course_id,
       name,
       TO CHAR(start_date, 'Month YYYY') AS formatted_date
FROM Courses;
```

Course_id	name	formatted_date
201	SQL Basics	May 2025
202	Python	June 2025

6. Create full date from parts using DATE_FROM_PARTS().

```
SELECT member_id,
        DATE_FROM_PARTS(start_year, start_month, start_day) AS full_start_date date
FROM Memberships;
```

member_id	full_start_date
1	2023-05-10
2	2022-11-25

7. Extend each renewal_date by 1 year using DATEADD().

```
SELECT sub_id,
        plan,
        DATEADD(YEAR, 1, renewal_date) AS extended_renewal_date
FROM Subscriptions;
```

Sub_id	plan	extended_renewal_date
11	Basic	2026-01-01
12	Premium	2026-03-15

8. Show current date and difference from order_date. Use current_Date and DateDiff().

```
SELECT order_id,
        order_date,
        CURRENT_DATE AS today_date,
        DATEDIFF(CURRENT_DATE, order_date) AS days_since_order
FROM orders;
```

order_id	order_date	today_date	days_since_order
1001	2025-04-15	2026-04-17	367
1002	2025-04-10	2026-04-17	372

9. Extract the year from training_date using DATE_PART() or EXTRACT().

```
SELECT training_id,
       topic,
       DATE_PART('Year', training_date) AS training_year
FROM Trainings;
```

training_id	topic	training_year
1	Safety	2025
2	Compliance	2025

10. Extract hour and minute from published_on.

```
SELECT post_id,
       title,
       EXTRACT(HOUR FROM published_on) AS hour_published,
       EXTRACT(MINUTE FROM published_on) AS minute_published
FROM Blog_posts;
```

Post_id	title	hour_published	minute_published
1	SQL tips	10	15
2	Data cleaning	16	45

11. Calculate days left until license expiry using DATEDIFF() and today's date

```
SELECT driver_id,
       license_expiry,
       DATEDIFF(license_expiry, CURRENT_DATE()) AS days_left
FROM Drivers;
```

driver_id	license_expiry	days_left
301	2025-08-10	-250
302	2023-12-31	-838

12. Display the current timestamp and calculate seconds since the message was sent.

```
SELECT message_id,
       sent_timestamp,
       CURRENT_TIMESTAMP() AS current_timestamp,
       TIMESTAMPDIFF(SECOND, sent_timestamp, CURRENTTIMESTAMP()) AS
       seconds_since_sent
FROM Messages;
```

message_id	sent_timestamp	current_timestamp	seconds_since_sent
1	2025-04-19 09:32:45	2026-04-17 08:00:00	31462035
2	2025-04-18 23:59:59	2026:04-17 08:00:00	31533601

13. Add 15 days to return_date using DATEADD() to show restock_date

```
SELEET return_id,
       return_date,
       DATEADD(DAY, 15, return_date) AS restock_date
FROM Returns;
```

return_id	return_date	restock_date
901	2025-04-05	2025-04-20
902	2025-04-01	2025-04-16

14. Convert assigned_on to date using TO_DATE() (if its stored as string)

```
SELECT assign_id,
        TO_DATE (assigned_on, 'YYYY-MM-dd') As assigned_on_date
FROM Assignments;
```

assign_id	assigned_on_date
1	2025-03-01
2	2025-03-05

15. Convert scheduled_time to formatted string like 'April 19, 2025 at 2:00 PM' using TO_CHAR().

```
SELECT meeting_id,
        TO_CHAR(scheduled_time, 'MMMM dd, yyyy "at" h:mm a') As
        formatted_time_meeting
FROM Meetings
```

meeting_id	formatted_meeting_time
1	April 19, 2025 at 2:00 PM
2	April 19, 2025 at 9:30 AM